LOGNORMAL DISTRIBUTION AND CONTINUOUS COMPOUNDING

Learning Outcome

- explain the relationship between normal and lognormal distributions and why the lognormal distribution is used to model asset prices when using continuously compounded asset returns

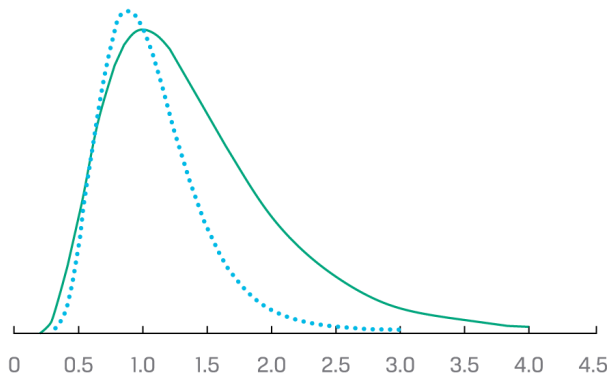
The Lognormal Distribution

Closely related to the normal distribution, the lognormal distribution is widely used for modeling the probability distribution of share and other asset prices. For example, the lognormal distribution appears in the Black–Scholes–Merton option pricing model. The Black–Scholes–Merton model assumes that the price of the asset underlying the option is lognormally distributed.

A random variable Y follows a lognormal distribution if its natural logarithm, ln Y, is normally distributed. The reverse is also true: If the natural logarithm of random variable Y, ln Y, is normally distributed, then Y follows a lognormal distribution. If you think of the term lognormal as "the log is normal," you will have no trouble remembering this relationship.

The two most noteworthy observations about the lognormal distribution are that it is bounded below by 0 and it is skewed to the right (it has a long right tail). Note these two properties in the graphs of the probability density functions (pdfs) of two lognormal distributions in Exhibit 1. Asset prices are bounded from below by 0. In practice, the lognormal distribution has been found to be a usefully accurate description of the distribution of prices for many financial assets. However, the normal distribution is often a good approximation for returns. For this reason, both distributions are very important for finance professionals.

Exhibit 1: Two Lognormal Distributions

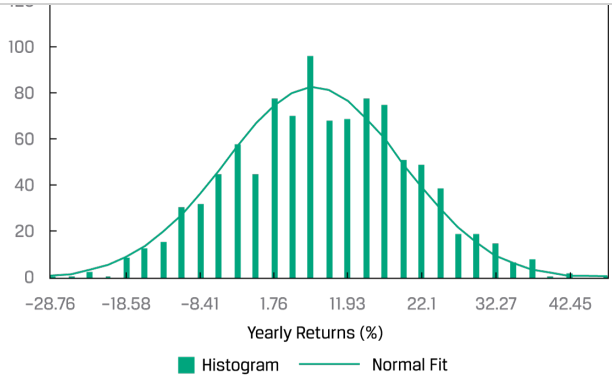


▶ Expand / collapse Extended Description

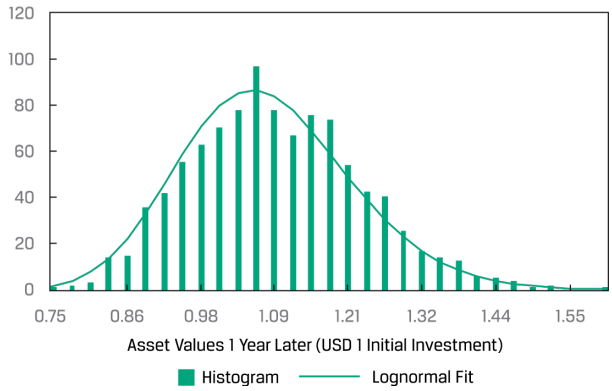
Like the normal distribution, the lognormal distribution is completely described by two parameters. Unlike many other distributions, a lognormal distribution is defined in terms of the parameters of a *different* distribution. The two parameters of a lognormal distribution are the mean and standard deviation (or variance) of its associated normal distribution: the mean and variance of ln Y, given that Y is lognormal. So, we must keep track of two sets of means and standard deviations (or variances): (1) the mean and standard deviation (or variance) of the associated normal distribution (these are the parameters) and (2) the mean and standard deviation (or variance) of the lognormal variable itself.

To illustrate this relationship, we simulated 1,000 scenarios of yearly asset returns, assuming that returns are normally distributed with 7 percent mean and 12 percent standard deviation. For each scenario i, we converted the simulated continuously compounded returns (r_i) to future asset prices with the formula Price(1 year later)$_i$ = USD1 × exp(r_i), where exp is the exponential function and assuming that the asset's price is USD1 today. In Exhibit 2, Panel A shows the distribution of the simulated returns together with the fitted normal pdf, whereas Panel B shows the distribution of the corresponding future asset prices together with the fitted lognormal pdf. Again, note that the lognormal distribution of future asset prices is bounded below by 0 and has a long right tail.

Exhibit 2: Simulated Returns (Normal PDF) and Asset Prices (Lognormal PDF)



B. Lognormal PDF



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The expressions for the mean and variance of the lognormal variable are challenging. Suppose a normal random variable X has expected value μ and variance σ^2. Define $Y = \exp(X)$. Remember that the operation indicated by $\exp(X)$ or e^X (where $e \approx 2.7183$) is the opposite operation from taking logs. Because $\ln Y = \ln[\exp(X)] = X$ is normal (we assume X is normal), Y is lognormal. What is the expected value of $Y = \exp(X)$? A guess might be that the expected value of Y is $\exp(\mu)$. The expected value is actually $\exp(\mu + 0.50\sigma^2)$, which is larger than $\exp(\mu)$ by a factor of $\exp(0.50\,\sigma^2) > 1$. To get some insight into this concept, think of what happens if we increase σ^2. The distribution spreads out; it can spread upward, but it cannot spread downward past 0. As a result, the center of its distribution is pushed to the right: The distribution's mean increases.

The expressions for the mean and variance of a lognormal variable are summarized below, where μ and σ^2 are the mean and variance of the associated normal distribution (refer to these expressions as needed, rather than memorizing them):

- Mean (μ_L) of a lognormal random variable = $\exp(\mu + 0.50\sigma^2)$.
- Variance (σ_L^2) of a lognormal random variable = $\exp(2\mu + \sigma^2) \times [\exp(\sigma^2) - 1]$.

Continuously Compounded Rates of Return

We now explore the relationship between the distribution of stock return and stock price. In this section, we show that if a stock's continuously compounded return is normally distributed, then future stock price is necessarily lognormally distributed. Furthermore, we show that stock price may be well described by the lognormal distribution even when continuously compounded returns do not follow a normal distribution. These results provide the theoretical foundation for using the lognormal distribution to model asset prices.

Showing that the stock price at some future time T, P_T, equals the current stock price, P_0, multiplied by e raised to power $r_{0,T}$, the continuously compounded return from 0 to T:

$$P_T = P_0 \exp(r_{0,T}).$$

We showed in an earlier lesson that $r_{0,T}$, the continuously compounded return to time T, is the sum of the one-period continuously compounded returns, as follows:

$$r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \ldots + r_{0,1}. \qquad 1$$

If these shorter-period returns are normally distributed, then $r_{0,T}$ is normally distributed (given certain assumptions) or approximately normally distributed (not making those assumptions). As P_T is proportional to the log of a normal random variable, P_T is lognormal.

A key assumption in many investment applications is that returns are **independently and identically distributed** (i.i.d.). Independence captures the proposition that investors cannot predict future returns using

mean μ and variance σ^2 (but making no normality or other distributional assumption). Then,

$$E(r_{0,T}) = E(r_{T-1,T}) + E(r_{T-2,T-1}) + \ldots + E(r_{0,1}) = \mu T, \quad (2)$$

(we add up μ for a total of T times), and

$$\sigma^2(r_{0,T}) = \sigma^2 T \quad (3)$$

(as a consequence of the independence assumption).

The variance of the T holding period continuously compounded return is T multiplied by the variance of the one-period continuously compounded return; also, $\sigma(r_{0,T}) = \sigma\sqrt{T}$. If the one-period continuously compounded returns on the right-hand side of Equation 1 are normally distributed, then the T holding period continuously compounded return, $r_{0,T}$, is also normally distributed with mean μT and variance $\sigma^2 T$. This is because a linear combination of normal random variables is also a normal random variable.

Even if the one-period continuously compounded returns are not normal, their sum, $r_{0,T}$, is approximately normal according to the central limit theorem. Now compare $P_T = P_0\exp(r_{0,T})$ to $Y = \exp(X)$, where X is *normal* and Y is lognormal (as we discussed previously). Clearly, we can model future stock price P_T as a lognormal random variable because $r_{0,T}$ should be at least approximately *normal*. This assumption of normally distributed returns is the basis in theory for the lognormal distribution as a model for the distribution of prices of shares and other financial assets.

Continuously compounded returns play a role in many asset pricing models, as well as in risk management. **Volatility** measures the standard deviation of the continuously compounded returns on the underlying asset; by convention, it is stated as an annualized measure. In practice, we often estimate volatility using a historical series of continuously compounded daily returns. We gather a set of daily holding period returns, convert them into continuously compounded daily returns and then compute the standard deviation of the continuously compounded daily returns and annualize that number using Equation 3.

Annualizing is typically done based on 250 days in a year, the approximate number of business days that financial markets are typically open for trading. Thus, if daily volatility were 0.01, we would state volatility (on an annual basis) as $0.01\sqrt{250} = 0.1581$. Example 1 illustrates the estimation of volatility for the shares of Astra International.

EXAMPLE 1

Volatility of Share Price

Suppose you are researching Astra International (Indonesia Stock Exchange: ASII) and are interested in Astra's price action in a week in which international economic news had significantly affected the Indonesian stock market. You decide to use volatility as a measure of the variability of Astra shares during that week. Exhibit 3 shows closing prices during that week.

Exhibit 3: Astra International Daily Closing Prices

Day	Closing Price (Indonesian rupiah, IDR)
Monday	6,950
Tuesday	7,000
Wednesday	6,850
Thursday	6,600
Friday	6,350

Use the data provided to do the following:

1. Estimate the volatility of Astra shares. (Annualize volatility on the basis of 250 trading days in a year.)

Hide Solution

Solution:

First, calculate the continuously compounded daily returns; then, find their standard deviation in the usual way. In calculating sample variance, to get sample standard deviation, the divisor is sample size minus 1.

ln(7,000/6,950) = 0.007168.

ln(6,850/7,000) = −0.021661.

ln(6,600/6,850) = −0.037179.

ln(6,350/6,600) = −0.038615.

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Variance = 0.000452.

Standard deviation = 0.021261.

The standard deviation of continuously compounded daily returns is 0.021261. Equation 3 states that $\hat{\sigma}(r_{0,T}) = \hat{\sigma}\sqrt{T}$. In this example, $\hat{\sigma}$ is the sample standard deviation of one-period continuously compounded returns. Thus, $\hat{\sigma}$ refers to 0.021261. We want to annualize, so the horizon T corresponds to one year. Because $\hat{\sigma}$ is in days, we set T equal to the number of trading days in a year (250).

Therefore, we find that annualized volatility for Astra stock that week was 33.6 percent, calculated as $0.021261\sqrt{250} = 0.336165$.

2. Calculate an estimate of the expected continuously compounded annual return for Astra.

Hide Solution

Solution:

Note that the sample mean, −0.022572 (from the Solution to 1), is a sample estimate of the mean, μ, of the continuously compounded one-period or daily returns. The sample mean can be translated into an estimate of the expected continuously compounded annual return using Equation 2, $\hat{\mu}T = -0.022572(250)$ (using 250 to be consistent with the calculation of volatility).

3. Discuss why it may not be prudent to use the sample mean daily return to estimate the expected continuously compounded annual return for Astra.

Hide Solution

Solution:

Four daily return observations are far too few to estimate expected returns. Further, the variability in the daily returns overwhelms any information about expected return in a series this short.

4. Identify the probability distribution for Astra share prices if continuously compounded daily returns follow the normal distribution.

Hide Solution

Solution:

Astra share prices should follow the lognormal distribution if the continuously compounded daily returns on Astra shares follow the normal distribution.

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We have shown that the distribution of stock price is lognormal, given certain assumptions. Earlier we gave bullet-point expressions for the mean and variance of a lognormal random variable. In the context of a stock price, the $\hat{\mu}$ and $\hat{\sigma}^2$ in these expressions would refer to the mean and variance of the T horizon, not the one-period, continuously compounded returns compatible with the horizon of P_T.

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SC Hi all, could you help explain graph 2? What is that Asset values 1 year later? Thank you,

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MN where am i going wrong when computing the variance (i am finding a sample variance of 0.0034 and standard deviation of 0.0583)

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In example 1, question 1. What if we had weekly, monthly or semi-annual closing prices, how are we going to annualize that?

Tags: lognormaldistro, annualizingrate — 1 reply | Last Activity: 5 months ago

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